PRAJJAWAL SINGH

singhprajjawal87@gmail.com | +91 8303159391 | LinkedIn | GitHub | LeetCode | CodeChef | Codeforces

SUMMARY

Backend Engineer with 1+ year of experience building scalable, production-grade backend systems and APIs with FastAPI, Django and Java/Spring Boot, alongside deploying cloud-hosted services and real-time data pipelines. Hands-on with REST API design, third-party integrations (webhooks, idempotency, retries), PostgreSQL/MySQL, Redis, Kafka, ClickHouse, AWS, Docker, Kubernetes (Helm, Ingress) and Nginx. Strong in DSA and CS fundamentals (1,000+ problems solved); LeetCode Guardian (rating 2130, top 1.92%). Daily user of AI coding tools (Claude Code, Cursor, Copilot) to ship quality code faster.

TECHNICAL SKILLS

Languages: Java (17+), Python, C++, SQL, JavaScript

Java & Frameworks: Spring Boot, Spring MVC, Dependency Injection, Hibernate/JPA, MapStruct, Liquibase/Flyway, JUnit

Backend & APIs: REST API Design, Django, FastAPI, GraphQL, OAuth2, JWT, Authentication & Authorization, Microservices

Third-Party Integrations: Webhooks / Callback Processing, Idempotency Keys, Retries, Reconciliation, Replay Handling, HMAC Signing, JWS/JWE, Payment Gateway & Partner API Integration

Databases & Caching: PostgreSQL, MySQL, Indexing & Query Optimisation (N+1), ClickHouse, ScyllaDB, MongoDB, Redis, Memcached

Messaging & Architecture: Apache Kafka, SQS, RabbitMQ, Event-Driven Architecture, Eventual Consistency, CQRS, Event Sourcing, Distributed Systems, Stream Processing, DB Replication, Scalability, Fault Tolerance

Cloud & DevOps: AWS (EKS, RDS, S3, Secrets Manager), CDN, Load Balancer, Docker, Kubernetes, Helm, Ingress, Nginx (reverse proxy), CI/CD, GitHub Actions

Observability: OpenTelemetry, SigNoz, Grafana, Structured Logging, Distributed Tracing, Metrics

OS / Tools: Linux, Shell, Git, GitHub, Claude Code, Cursor, GitHub Copilot

CS Fundamentals: DSA, OOP, Concurrency, HTTP, JSON, System Design (URL Shortener, Rate Limiter)

Design Patterns: Repository, Factory, Strategy, Observer, Singleton, MVC, CQRS, State Machine

EXPERIENCE

Backend Engineer - TailNode (Full-time) *Delhi, India | Jul 2025 - Present*

- Integrated a CDN layer that cut static-asset delivery costs by 90% and reduced global page-load times by 60% for 50K+ active users, improving retention and user experience.
- Containerized backend services with Docker and deployed them on Kubernetes using Helm charts and Ingress routing behind a load balancer, supporting a 99.9% uptime SLA.
- Configured Nginx as a reverse proxy and load balancer for backend services in a distributed environment handling high-volume, real-time data streams.
- Automated the user-data ETL pipeline (Kafka to ClickHouse) syncing and transforming 500K+ daily events, shrinking reporting latency from hours to under 2 minutes; added data replication for high availability and fault tolerance, with CQRS and Event Sourcing for audit-ready data models and eventual-consistency handling.
- Developed low-latency REST and FastAPI endpoints (sub-50ms p99) secured via OAuth2/JWT, applying Repository, Factory and Strategy patterns and dependency-injection-style layering for maintainable service architecture.
- Implemented the Ramer-Douglas-Peucker (RDP) algorithm, reducing GPS coordinate payloads by ~90% while preserving spatial accuracy for faster map rendering.
- Led system design reviews covering PostgreSQL sharding and indexing, Redis caching, ScyllaDB write optimisation and distributed consistency (CAP theorem trade-offs).
- Debugged production issues across logs, metrics, database and service responses, and used AI coding tools (Claude Code, Cursor, Copilot) with critical code review to ship tested changes faster.

PROJECTS

Social Media Application - Skyline | JavaScript, Node.js, React, MongoDB | GitHub

- Built a full-stack social platform with auth, posts and feeds; REST APIs integrated with a responsive React frontend, deployed with live hosting.
- Cut average feed query time by 70% with MongoDB compound indexes; implemented CI/CD via GitHub Actions.
- Applied MVC and Observer design patterns for a clean, scalable codebase serving 1,000+ concurrent users.

Quiz Competition Platform | JavaScript, Redis, Django, React | GitHub

- Delivered real-time leaderboard updates via Redis pub/sub (<100ms latency) for 500+ concurrent participants.
- Reduced page load by 45% through lazy loading, code splitting and Django query optimisation.
- Used the State Machine pattern for quiz session lifecycle management, ensuring data integrity.

EDUCATION

National Institute of Technology, Tiruchirappalli | B.Tech, Electronics & Communication Engineering *2021 - 2025*

CGPA: 7.91 / 10 | Don Bosco School | Class XII (PCM) - 91.8%

ACHIEVEMENTS

- LeetCode Guardian: rating 2130, top 1.92% globally; 953 problems solved (Easy 234 / Medium 490 / Hard 229); 330+ day streak, 198 active days and max streak of 41 in the past year.
- CodeChef 4-star, peak rating 1922; Codeforces Specialist, max rating ~1550; 1,000+ DSA problems solved across competitive platforms.